# Local Library Archive Legacy Migration

## Team Portfolio Report

**Module:** 5COM2006
**Project Scenario:** Option 1
**Team Members:** Jesse Boateng, Ma Yuan
**Submission:** Team Portfolio

## Project Overview

For this project, we developed a legacy migration solution for the Local Library Archive (LLA). The purpose of the system was to replace the library's unstructured mix of Excel spreadsheets and physical folders with a more reliable digital archive system. The final solution was designed around a database-backed internal web platform that can store and organise books, authors, member details, and loan history in a structured way.

The system was designed using a two-tier architecture in AWS, with a web server hosting the internal search interface and a separate database server storing the application data. The main focus of the project was data integrity, meaning the system had to reduce duplication, preserve consistency, and handle member information responsibly in line with GDPR principles. To support operational reliability, the project also included automation through backup and log rotation scripts.

## Software Used

The main software used in this project was Ubuntu Server, Apache, PHP, MySQL, Bash, and cron. GitHub was used to store technical artefacts and scripts, while Trello was used to manage tasks and maintain the Kanban workflow. For database implementation, SQL was used to define the schema and seed the initial records. The overall system was designed as a LAMP-style deployment suitable for an internal library archive system.

## Project Plan

The project was planned throughout the term using a Gantt chart to organise responsibilities, tasks, and overall progress. The work began with planning and requirements analysis, followed by Trello task allocation and repository organisation. The implementation stage included AWS web and database server deployment, security group configuration, MySQL setup, database design, integration testing, search interface support, and automation scripting. After the main infrastructure and schema were prepared, the project moved into documentation, diagram preparation, evidence gathering, and final portfolio assembly for submission.

**Legacy Migration Project Gantt Chart**

| Task | Owner | Status | Week 3 | Week 4 | Week 5 | Week 6 | Week 7 | Week 8 | Week 9 | Week 10 | Week 11 | Week 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Project planning and requirements | Yuan | Completed | | | | | | | | | | |
| Trello board setup and user stories | Jesse | Completed | | | | | | | | | | |
| Use Case diagram | Jesse | Completed | | | | | | | | | | |
| ER diagram | Yuan | Completed | | | | | | | | | | |
| Gantt chart creation | Jesse | Completed | | | | | | | | | | |
| AWS web server deployment | Jesse | Completed | | | | | | | | | | |
| AWS database server deployment | Yuan | Completed | | | | | | | | | | |
| Security group configuration | Jesse | Completed | | | | | | | | | | |
| MySQL setup and library database creation | Yuan | Completed | | | | | | | | | | |
| Web-to-database integration test | Jesse | Completed | | | | | | | | | | |
| Library search interface setup | Yuan | Completed | | | | | | | | | | |
| Nightly backup automation (2 AM) | Yuan | Completed | | | | | | | | | | |
| Privacy controls and server hardening | Jesse | Completed | | | | | | | | | | |
| SQL script creation and testing | Yuan | Completed | | | | | | | | | | |
| Backup and security testing | Yuan | Completed | | | | | | | | | | |
| Documentation and GitHub uploads | Jesse | Completed | | | | | | | | | | |
| Final report preparation | Jesse | Completed | | | | | | | | | | |
| Final submission packaging | Jesse | Completed | | | | | | | | | | |

# System Design

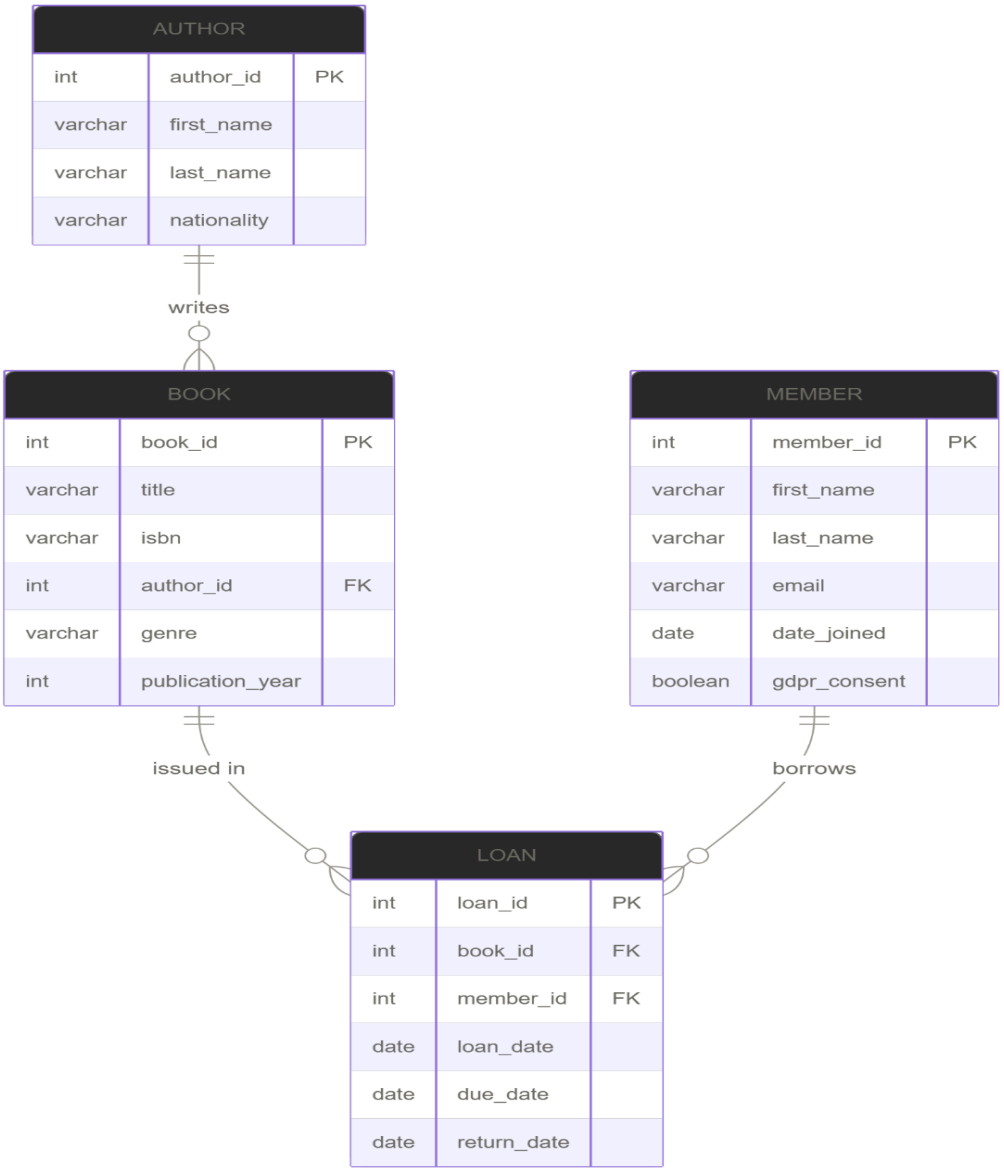
The system was designed as a simple but secure two-tier architecture. The web tier hosts the internal archive search interface and handles requests from staff, while the database tier stores structured data relating to books, authors, members, and loan history. This separation improves maintainability and reduces unnecessary exposure of the data layer. The overall design is represented using an Entity Relationship Diagram (ERD), a Network Topology diagram, and a Use Case diagram.

# Entity Relationship Diagram

The ERD models the main data required by the Local Library Archive system. The key entities are Books, Authors, Members, Loans, and the linking table Book_Authors. This design supports a normalised database structure by separating repeated data into related tables and reducing duplication.

Because books can have multiple authors and authors can write multiple books, a junction table was used to model the many-to-many relationship between Books and Authors. Loan records connect individual members to the books they borrow, making it possible to store and track borrowing history clearly. This structure supports the key challenge of the project by helping prevent duplicate entries and improve data integrity.

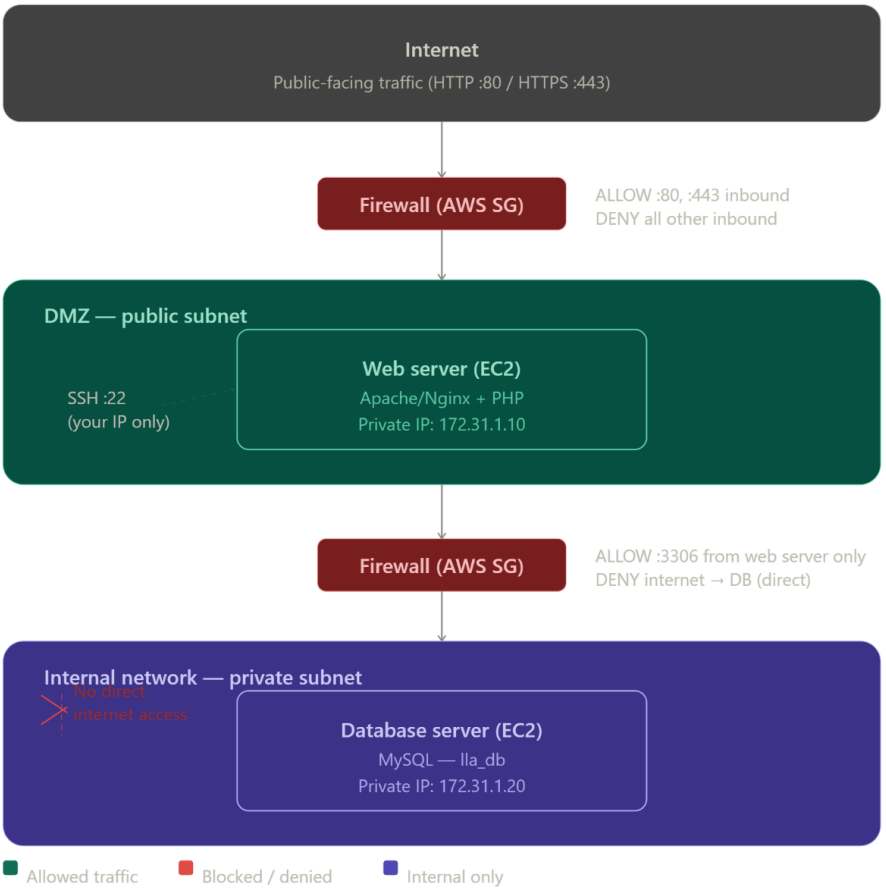
To strengthen the design further, unique constraints were placed on important fields such as ISBN and member email addresses. These decisions help reduce duplicate records and improve the consistency of the archive system.



# Network Topology

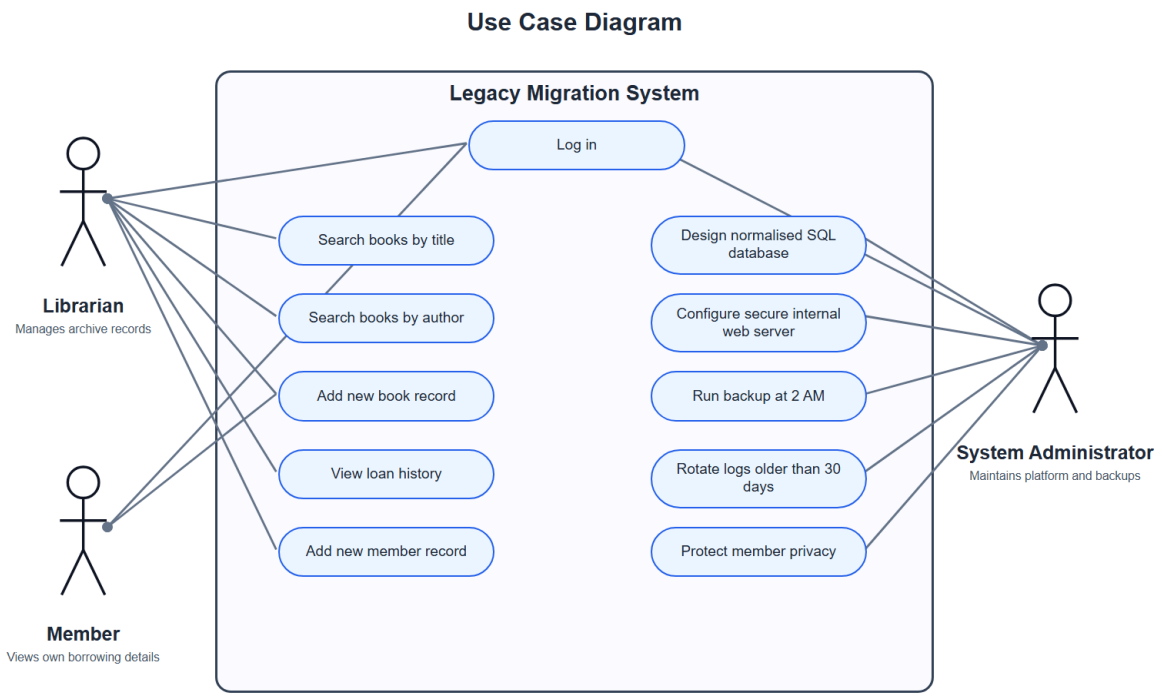
The network topology was designed to separate the application and database services into different AWS EC2 instances. The web server was configured to host the internal archive interface and allow controlled access over HTTP and HTTPS, while administrative SSH access was restricted. The database server ran MySQL and was not intended to be directly exposed for public application traffic.

Communication between the web server and database server was controlled through AWS security group rules, with MySQL traffic on port 3306 only permitted from the web tier. The database server was designed to remain protected from direct public access, which supports both security and privacy requirements for the library data.



# Use Case Diagram

An additional Use Case diagram was created to show the interaction between the main actors in the system and the archive functions. The primary actors are the Librarian, Member, and System Administrator. The Librarian interacts with the system to search books, manage archive records, and view loan history. Members are represented in relation to their stored records and borrowing activity. The System Administrator is responsible for maintaining the internal web server, supporting the database environment, and running the backup and maintenance routines required by the project.




# Testing Evidence

Testing was included to evaluate the reliability, integrity, and operational readiness of the Legacy Migration environment. Our work focused on verifying database structure, confirming communication between the web and database tiers, and ensuring that the automation scripts supported the backup and maintenance requirements of the system. A summary of these activities is provided below.

# Database and Integration Testing Summary

The database schema was reviewed to ensure it satisfied normalisation requirements and reduced duplication. The structure was designed in Third Normal Form (3NF), and specific constraints such as unique ISBN values and unique member email addresses were used to support data integrity. Indexes were also introduced on important search-related fields to improve query performance.

Integration testing also formed an important part of the project. The connection between the web tier and the database tier was checked to ensure that the internal search interface could communicate with MySQL successfully. An early issue occurred because the database was initially listening only on localhost. After this was corrected so that the database listened on the appropriate interface, communication between the web and database layers worked as expected.

## Automation and Maintenance Testing Summary

A key requirement of the project was automation. To address this, backup and maintenance scripts were created to support regular database protection and file management. The backup script was designed to run nightly at 2 AM and used `mysqldump` to produce compressed backups of the database. The log rotation and maintenance script was designed to remove old files after 30 days, helping control storage use and support long-term maintainability.

These scripts were designed to be safe and repeatable, and care was taken to make them suitable for practical use in a small internal system. This automation supported both reliability and operational discipline, which are important in a migration project where archive records need to be preserved consistently.

## Security and Privacy Considerations

Security and privacy were important throughout the design of the system because member information forms part of the stored dataset. To support GDPR principles, the schema was designed to minimise unnecessary personal data and store only the fields required for the intended use of the system. Access to the database was restricted through security group rules, and the database application user was created according to the principle of least privilege.

The server environment also followed secure deployment practices by separating the application tier from the data tier and avoiding unnecessary public exposure of the database server. These controls supported the project's aim of creating a digital archive system that is more secure and manageable than the original spreadsheet and paper-based approach.

## Team Contributions

Team contributions were divided across technical implementation and portfolio preparation. Yuan focused on the core technical development of the Legacy Migration solution, including the database design, SQL schema, automation scripts, and cloud infrastructure configuration. His work included the creation of the normalised schema, backup and maintenance scripting, and the AWS server setup required to support the application and database environment.

Jesse focused on project organisation, documentation, design artefacts, and final portfolio preparation. This included Trello management, GitHub organisation, diagram preparation

and formatting, report assembly, and overall submission packaging. Both members contributed to reviewing the final submission and ensuring that the portfolio matched the Local Library Archive project brief.

# Conclusion

In conclusion, our team produced a Legacy Migration solution for the Local Library Archive using AWS, Apache, PHP, MySQL, SQL, and Bash scripting. The project combined cloud deployment, data modelling, automation, and documentation into a structured digital archive solution that addresses the client’s need to move away from fragmented spreadsheet and paper-based records.